25900200 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCSTA101 - DATA ANALYSIS USING JAMOVI AND INTRODUCTION TO R

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **2** marks

1. If a dataset contains the values 2, 4, 6, 8, and 10, what is the mean? [U] / [CO2]

   a). 8

   b). 6

   c). 30

   d). 5

2. For a dataset, variance = 0. What is the standard deviation of the data? [K] / [CO2]

   a). 1

   b). Not defined

   c). 0

   d). None of the above

3. The difference between 25°C and 30°C is meaningful and consistent, regardless of the zero point. This indicates which scale of measurement? [An] / [CO2]

   a). Nominal

   b). Ordinal

   c). Interval

   d). Ratio

4. Which of the following statements about quartile deviation is true? [K] / [CO2]

   a). It is sensitive to outliers

   b). It is not affected by outliers

   c). It is a measure of central tendency

   d). None of the above

5. Find the mode of the scores: 4,4,6,6,2,3,3,3,3,3,4,5,6,5,6,6,3. [An] / [CO2]

   a). 3

   b). 6

   c). 4

   d). 5

6. In a regression model, the equation is y=21-5x. What can be inferred about the relationship between x and y? [U] / [CO3]

   a). As x decreases y also decreases

   b). x and y are unrelated

   c). As x increases y also increases

   d). As x increases y decreases

7. Which test is used to check for normality of residuals in linear regression? [K] / [CO3]

   a). Shapiro-Wilk test

   b). Levene's test

c). Durbin-Watson test

d). All of the above

8. What will be the output of the following code? [A] / [CO4]
x<- 1 while (x<=5) { print (x) x<- x+1 }

a). Error

b). No output

c). 1 2 3 4 5

d). 2 3 4 5

9. For the vector x=(10,15,20,25,30,35), what will be the output of the code x[-1]? [A] / [CO4]

a). 10,15,20,25,30

b). 15,20,25,30,35

c). 10

d). None of the above

10. What is R Markdown primarily used for? [K] / [CO4]

a). Writing HTML code

b). Managing databases

c). Combining R code, text, and plots in a document

d). Running java programs

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **3** questions

Each question carries **5** marks

11. Define median. Calculate the median for the following dataset of exam scores from a class test of 9 students:23,18,25,19,21,22,16,24,20. [An] / [CO2]

12. Define the term dispersion. Explain variance as a measure of dispersion. [U] / [CO2]

13. Define correlation. Also explain the concept of correlation coefficient and how it is interpreted? [A] / [CO3]

14. Distinguish between comparison and assignment operators in R. [U] / [CO4]

15. Define a dataframe. Suppose we have three vectors: x = (2,1,3,7), y = (4,2,3,5) and z = (10,20).
    Write the R script to create the following data frames: a) (x, y,z) b) (x, y) c) (x, z).
    Also, write the output obtained. [A] / [CO4]

[5x3 = 15]